WORKED SOLUTION

STEP II 1989, Question 7

Pure • Second-Order Differential Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q7

The key idea

The terms $x\,y''$ and $-b\,y'$ look awkward together, but a change of independent variable $t = x^{\alpha}$ produces a first-derivative term from y'' that can be made to cancel $-b\,y'$ exactly. Choosing α to do this leaves simple harmonic motion in t. After that, the behaviour as $x \to 0$ comes down to how the factor x^b in $\mathrm{d}y/\mathrm{d}x$ behaves, and that depends on the sign of b.

Solution

Changing the variable

Put $t = x^{\alpha}$, with α to be chosen, and regard y as a function of t. By the chain rule, with $\dot{y} = \mathrm{d}y/\mathrm{d}t$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \alpha x^{\alpha-1}\dot{y},$$
$$\frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \alpha(\alpha-1)x^{\alpha-2}\dot{y} + \alpha^2 x^{2\alpha-2}\ddot{y},$$

where the second line uses the product rule and $\frac{\mathrm{d}}{\mathrm{d}x}\dot{y} = \alpha x^{\alpha-1}\ddot{y}$. Substituting,

$$x\frac{\mathrm{d}^2y}{\mathrm{d}x^2} - b\frac{\mathrm{d}y}{\mathrm{d}x} = \alpha^2 x^{2\alpha-1}\ddot{y} + \alpha(\alpha-1-b)\,x^{\alpha-1}\dot{y}.$$

The $\dot{y}$ term disappears if we choose $\alpha = b+1$, which is non-zero because $b > -1$. Then $2\alpha - 1 = 2b+1$, so the equation becomes

$$(b+1)^2 x^{2b+1}\ddot{y} + x^{2b+1}y = 0 \quad\Longrightarrow\quad \ddot{y} + \frac{y}{(b+1)^2} = 0,$$

dividing by $x^{2b+1} \neq 0$ (we have $x > 0$). This is simple harmonic motion in t with angular frequency $1/(b+1)$, so

$$\boxed{y = A\cos\left(\frac{x^{b+1}}{b+1}\right) + B\sin\left(\frac{x^{b+1}}{b+1}\right)}$$

with A, B arbitrary constants. Two constants for a second-order equation: this is the general solution.

Behaviour as $x \to 0$

Write $u = x^{b+1}/(b+1)$. Since $b+1 > 0$, $u \to 0^+$ as $x \to 0^+$, so

$$y \to A\cos 0 + B\sin 0 = A.$$

Hence $y \to 1$ forces $A = 1$. Differentiating, using $\mathrm{d}u/\mathrm{d}x = x^b$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = x^b(-\sin u + B\cos u) \qquad (A = 1).$$

Case $b > 0$. Now $x^b \to 0$, while the bracket tends to B, which is finite. So $\mathrm{d}y/\mathrm{d}x \to 0$ *whatever the value of* B. Every member of the family

$$y = \cos\left(\frac{x^{b+1}}{b+1}\right) + B\sin\left(\frac{x^{b+1}}{b+1}\right), \qquad B \in \mathbb{R},$$

satisfies both conditions. Solutions exist, but they are not unique, as required.

> **Ask yourself** For an ordinary second-order equation, a value of y and a value of y' fix the solution. Why does that fail here?

The uniqueness theorem needs the equation in the form $y'' = f(x, y, y')$ with f well behaved. Dividing by x gives $y'' = (b/x)\,y' - x^{2b}y$, and the coefficient b/x blows up at $x = 0$. The point $x = 0$ is *singular*, so the usual guarantee doesn't apply there.

Case $b = 0$. Now $x^b = 1$ and $\mathrm{d}y/\mathrm{d}x \to B$. The condition forces $B = 0$, giving the unique solution $y = \cos x$. (This is just $y'' + y = 0$.)

Case $-1 < b < 0$. Now $x^b \to \infty$. If $B \neq 0$ then $Bx^b\cos u \to \pm\infty$, while the other term is small: since $\sin u \leq u$,

$$|x^b \sin u| \leq x^b \cdot \frac{x^{b+1}}{b+1} = \frac{x^{2b+1}}{b+1}.$$

More precisely $\sin u/u \to 1$, so $x^b \sin u \sim x^{2b+1}/(b+1)$. So we need $B = 0$, and then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -x^b \sin u \sim -\frac{x^{2b+1}}{b+1}.$$

This tends to 0 if and only if $2b+1 > 0$, that is $b > -\frac{1}{2}$. At $b = -\frac{1}{2}$ it tends to -2, and for $b < -\frac{1}{2}$ it is unbounded. So for $-\frac{1}{2} < b < 0$ there is exactly one solution, $y = \cos(x^{b+1}/(b+1))$, and for $-1 < b \leq -\frac{1}{2}$ there is none.

Conclusion

The conditions determine a unique solution precisely when

$$-\tfrac{1}{2} < b \leq 0,$$

and that solution is $y = \cos(x^{b+1}/(b+1))$.



Checks

- $b = 0$ gives $y'' + y = 0$, whose solutions $A\cos x + B\sin x$ match the formula.
- Numerically, the general solution satisfies the equation to within rounding (finite differences at random x, b, A, B). With $b = -\frac{1}{4}$, $B = 0$ we found $y' \approx -4.2 \times 10^{-5}$ at $x = 10^{-9}$, tending to 0. With $b = -\frac{1}{2}$ the value stays at -2, and with $b = 1$, $B = 5$ it is 5×10^{-9} at $x = 10^{-9}$.

Common slips

- Stopping at "$b \leq 0$ gives uniqueness": for $-1 < b \leq -\frac{1}{2}$ even the cosine solution has $y' \not\to 0$, so there is no solution at all.
- Forgetting the product rule when differentiating $\alpha x^{\alpha-1}\dot{y}$ with respect to x. The extra term is exactly what cancels $-b\,y'$.

Beyond STEP

The point $x = 0$ is a *regular singular point* of this equation. Near such points, the Frobenius method looks for solutions $x^r \sum c_n x^n$, and the indicial equation here has roots $r = 0$ and $r = b + 1$. Which conditions at the singular point pin down a solution depends on these exponents, exactly as we saw. Boyce and DiPrima, *Elementary Differential Equations*, covers series solutions and regular singular points.

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